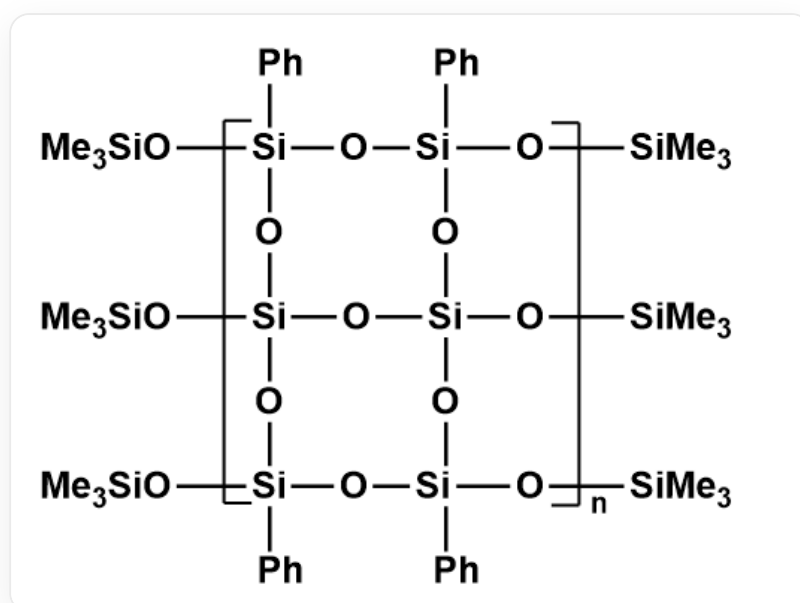


Question

The following figure shows a polymer with a multi-chain structure, which can be prepared as follows: Dissolve **A** ($\text{C}_6\text{H}_8\text{O}_3\text{Si}$) in dioxane, and gradually add a dioxane solution of **B** ($\text{C}_2\text{H}_6\text{O}_2\text{Cl}_2\text{Si}$) and triethylamine under argon protection to prepare monomer **C**; **C** undergoes dehydration under triethylamine catalysis to obtain polymer **D**; Dissolve **D** in THF, add a small amount of dilute hydrochloric acid, and stir thoroughly at room temperature for sufficient reaction; The resulting product undergoes dehydration again under triethylamine catalysis to obtain **E**; Finally, add a toluene solution of **F** to **E** for end-capping to obtain the final product.



This is a polymer. The repeating unit is $[\text{R1}][\text{Si}](\text{O}[\text{Si}]1(\text{O}[\text{R2}])\text{C}2=\text{CC}=\text{CC}=\text{C}2)(\text{O}[\text{Si}]([\text{R1}])(\text{O}3)\text{O}[\text{Si}]([\text{R1}])(\text{C}4=\text{CC}=\text{CC}=\text{C}4)\text{O}[\text{Si}](\text{O}[\text{R2}]) (\text{C}5=\text{CC}=\text{CC}=\text{C}5)\text{O}[\text{Si}]3(\text{O}[\text{R2}])\text{O}1)\text{C}6=\text{CC}=\text{CC}=\text{C}6$, R1 and R2 are the connection sites of the repeating unit; the terminal group R1 connected to Si is $-\text{OSiMe}_3$, and the terminal group R2 connected to O is $-\text{SiMe}_3$

Which of the following statements is incorrect?

A. **A** contains three hydroxyl groups.

B. **B** contains 2 methoxy groups.

C. **C** contains 8 **C**

D. **D** contains a 12-membered ring

E. The repeating unit in **E** contains 7 **O**.

Answer

Correct Answer: **C**

Detailed Explanation

Based on the product structure, **A** and **B** should be two small units of the polymer structure. Since **A** contains 6 C atoms, it can be determined that **A** is the part connected to the benzene ring, namely $\text{Si}(\text{OH})_3(\text{C}_6\text{H}_5)$, $\text{O}[\text{Si}](\text{O})(\text{O})\text{C1}=\text{CC}=\text{CC}=\text{C1}$. **A** contains three hydroxyl groups, so A is correct.

CHECKPOINT

1.5 PTS

A is $\text{O}[\text{Si}](\text{O})(\text{O})\text{C1}=\text{CC}=\text{CC}=\text{C1}$

B should be a structure not connected to the benzene ring. Combined with the molecular formula, it can be determined as $\text{Si}(\text{OCH}_3)_2\text{Cl}_2$, namely $\text{Cl}[\text{Si}](\text{Cl})(\text{OC})\text{OC}$. **B** contains 2 methoxy groups, so B is correct.

CHECKPOINT

1.5 PTS

B is $\text{Cl}[\text{Si}](\text{Cl})(\text{OC})\text{OC}$

The question indicates that **C** is a monomer obtained from the reaction of **A** and **B**, which is only the substitution of O for Cl, i.e., **C** is $\text{O}[\text{Si}](\text{O}[\text{Si}](\text{OC})(\text{OC})\text{O}[\text{Si}](\text{O})(\text{C1}=\text{CC}=\text{CC}=\text{C1})\text{O})(\text{O})\text{C2}=\text{CC}=\text{CC}=\text{C2}$. **C** contains 12 C atoms, so C is incorrect.

CHECKPOINT

1 PTS

C is O[Si](O[Si](OC)(OC)O[Si](O)(C1=CC=CC=C1)O)(O)C2=CC=CC=C2

From "**C** undergoes dehydration under triethylamine catalysis to obtain polymer **D**", it can be determined that hydroxyl groups are connected in this step, which is the polymerization process, i.e., the repeating unit of **D** is [R1][Si](O[Si]1(O[R2])C2=CC=CC=C2)(O[Si](OC)(OC)O[Si]([R1])(C3=CC=CC=C3)O[Si](O[R2])(C4=CC=CC=C4)O[Si](OC)(OC)O1)C5=CC=CC=C5, R1 and R2 are the connection sites of the repeating unit; the terminal group R1 near Si is 2 OH, and the terminal group R2 near O is 2 H. **D** contains a 12-membered ring, so D is correct.

CHECKPOINT

1 PTS

D's repeating unit is [R1][Si](O[Si]1(O[R2])C2=CC=CC=C2)(O[Si](OC)(OC)O[Si]([R1])(C3=CC=CC=C3)O[Si](O[R2])(C4=CC=CC=C4)O[Si](OC)(OC)O1)C5=CC=CC=C5, R1 and R2 are the connection sites of the repeating unit; the terminal group R1 near Si is 2 OH, and the terminal group R2 near O is 2 H

Subsequently, the addition of hydrochloric acid will hydrolyze the methoxy groups into hydroxyl groups, and then the hydroxyl groups formed by hydrolysis undergo dehydration polymerization. Therefore, the repeating unit of **E** is [R1][Si](O[Si]1(O[R2])C2=CC=CC=C2)(O[Si]([R1])(O3)O[Si]([R1])(C4=CC=CC=C4)O[Si](O[R2])(C5=CC=CC=C5)O[Si]3(O[R2])O1)C6=CC=CC=C6, R1 and R2 are the connection sites of the repeating unit; the terminal group R1 near Si is 3 OH, and the terminal group R2 near O is 3 H. The repeating unit with the smallest number of molecules in **E** contains 7 O atoms, so E is correct.

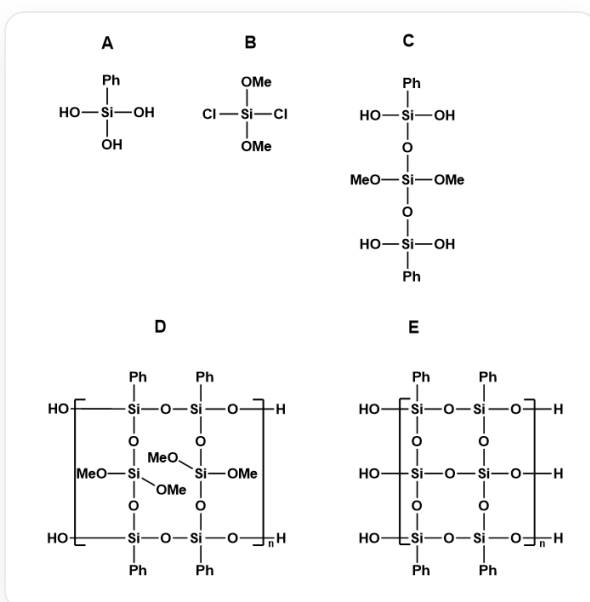
CHECKPOINT

1 PTS

E's repeating unit is $[R1][Si](O[Si]1(O[R2])C2=CC=CC=C2)(O[Si]([R1])(O3)O[Si]([R1])(C4=CC=CC=C4)O[Si](O[R2])(C5=CC=CC=C5)O[Si]3(O[R2])O1)C6=CC=CC=C6$, R1 and R2 are the connection sites of the repeating unit; the terminal group R1 near Si is 3 OH, and the terminal group R2 near O is 3 H

F is a capping agent and, based on the product structure, should be $SiMe_3X$, $X = Cl \cdot Br \cdot OTf$. Since it is not fixed, no further questions are asked.

In summary, choose option C.



Structures of **A** to **E**. **A** is $O[Si](O)(O)C1=CC=CC=C1$; **B** is $Cl[Si](Cl)(OC)OC$; **C** is $O[Si](O[Si](OC)(OC)O[Si](O)(C1=CC=CC=C1)O)(O)C2=CC=CC=C2$; The repeating unit of **D** is $[R1][Si](O[Si]1(O[R2])C2=CC=CC=C2)(O[Si](OC)(OC)O[Si]([R1])(C3=CC=CC=C3)O[Si](O[R2])(C4=CC=CC=C4)O[Si](OC)(OC)O1)C5=CC=CC=C5$, R1 and R2 are the connection sites of the repeating unit; the terminal group R1 near Si is 2 OH, and the terminal group R2 near O is 2 H; The repeating unit of **E** is $[R1][Si](O[Si]1(O[R2])C2=CC=CC=C2)(O[Si]([R1])(O3)O[Si]([R1])(C4=CC=CC=C4)O[Si](O[R2])(C5=CC=CC=C5)O[Si]3(O[R2])O1)C6=CC=CC=C6$, R1 and R2 are the connection sites of the repeating unit; the terminal group R1 near Si is 3 OH, and the terminal group R2 near O is 3 H